Marco Maida

Senior Machine Learning Engineer

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I'm a software and machine learning engineer with over **ten years of experience** delivering systems across **industrial software**, **videogames**, **real-time systems**, **formal verification**, and **autonomous vehicles**.
At **Wayve**, I develop real-time inference systems for autonomous vehicles. My work covers model and runtime co-design, heterogeneous accelerators, numerical validation, and latency optimisation. I work primarily in **C++**, **Rust**, and **Python**, with an emphasis on measurable performance, correctness, and production constraints.

Professional Experience

Wayve Ltd
London, United Kingdom

2026–
- **Senior Machine Learning Engineer**
  Co-design models and runtime architectures for real-time execution on resource-constrained automotive accelerators. Partition workloads across CPUs, GPUs, and NPUs; start model components as their sensor inputs become available; and maintain validation suites for numerical precision. Optimise end-to-end latency, throughput, and resource utilisation. *(C++, Rust, Python)*.

2024–2026
- **Tech Lead, Inference Performance**
  Led a team of four responsible for Wayve's on-board inference stack. Extended the platform across multiple inference frameworks and accelerators while supporting the same execution path in every vehicle and simulation environment. Owned latency, reliability, and technical delivery for the core inference component. *(C++, Rust, Python)*.

2022–2024
- **Senior Software Engineer**
  Joined a seven-person team building Wayve's second-generation on-board platform and shipping it to its first two vehicles. Built the initial real-time inference pipeline; Lidar, GNSS, and microphone stacks; data collection and upload features; and the vehicles' internal software network. *(ROS2, Linux, C++, Rust, Python)*.

34BigThings
Turin, Italy

2016–2019
- **Software Engineer – Videogames**
  Shipped five single-player and online multiplayer titles across Steam, PS4, Xbox One, Nintendo Switch, and mobile. Developed game infrastructure, AI, gameplay systems, and production tools in Unity and Unreal Engine. *(Unity3D, C#, Unreal Engine, C++)*.

Maserati / Choralia
Turin, Italy

2015–2016
- **Software Engineer – Games and Simulation**
  Delivered two B2B projects: a browser and mobile educational game, and a 3D product-visualisation tool. Led implementation and supervised an external artist. *(Unity3D, C#, JavaScript)*.

R.O. srl

2013–2016
- **Software Engineer – Industrial Software**
  Developed software for glass-processing factories covering order tracking, machine scheduling, waste reduction, and logistics. Managed four junior engineers during the third year. *(C, C++, C#, SQL)*.

Education and Research

2022 **Research Internship** Bloomberg LP
I worked on accelerating SAT solving using GPUs. *(C++, CUDA)*.

2019-2022 **PhD Student** Max Planck Institute
My research focused on formal verification and real-time systems, specifically verifying the timeliness of software systems. I published three papers and mentored three interns. *(COQ, C, Rust)*.

2019-2022 **Master's in Computer Science** Technische Universität Kaiserslautern
I specialized in real-time systems as part of a joint master's and PhD program.

2016-2019 **Bachelor's in Computer Science** Università degli Studi di Torino
I specialized in computability and formal methods.

Selected Publications

2025 **Claycode: Stylable and Deformable 2D Scannable Codes** SIGGRAPH 2025
★*Journal publication in Transactions of Graphics. Selected for CAF trailer.*
We introduced Claycodes, a new type of visual code that breaks free from traditional rigid grids like QR codes. Claycodes encode data as a tree structure and allow full customizability and deformation, with real-time scanning capabilities. (https://arxiv.org/abs/2505.08666).

2022 **From Intuition to Coq: A Case Study in Verified Response-Time Analysis, FIFO Scheduling** RTSS 2022
We developed a formally verified response-time analysis for FIFO schedulers, challenging traditional pen-and-paper methods. (https://people.mpi-sws.org/~kbedarka/rtss22.pdf).

2021 **Foundational Response-Time Analysis as Explainable Evidence of Timeliness** ECRTS 2022
★*Outstanding Paper Award.*
I developed POET, a tool for formally verified worst-case scenario timing analysis. (https://drops.dagstuhl.de/opus/volltexte/2022/16336/pdf/LIPIcs-ECRTS-2022-19.pdf).

Selected Open-Source Projects

2025 **Claycode** https://claycode.io
Inventor and lead contributor. Claycode includes a generator website, an Android scanner app, and a large amount of research code.

2020-2022 **PROSA** https://gitlab.mpi-sws.org/RT-PROOFS/rt-proofs
Main contributor to PROSA, one of the leading formally-verified frameworks in the real-time systems community.

2022 **Treecode** www.maida.me/treecode
The precursor to Claycode, a novel 2D scannable code that encodes messages as unique trees.

2021 **POET** https://gitlab.mpi-sws.org/RT-PROOFS/POET
First-ever implementation of a foundational response-time analysis tool, created as part of my first academic publication.

2018 **Fast Mobile Cycle (FMC) Framework and Toolkit** www.github.com/34OpenThings
Developed a Unity3D framework that accelerates production-ready casual game creation, paired with a Python toolkit for bulk operations.